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**Table 5.7 Abundance of important alkaline earth metals and their sources**

| Element | Abundance in Earth crust by weight (ppm) | Mineral source |
|---|---|---|
| Be | 2.0 | beryl $Be_3Al_2Si_6O_{18}$ |
| Mg | 27640 | Carnallite ($KCl.MgCl_2.6H_2O$) Dolomite $MgCO_3$ $CaCO_3$ |
| Ca | 1.84 | Fluorapatite $Ca_5(PO_4)_3$ F |
| Sr | 384 | Celestite $SrSO_4$ |
| Ba | 390 | barytes $BaSO_4$ |

### 5.5.1 General characteristics of alkaline earth metals

#### Physical state

Beryllium is rare and radium is the rarest of all comprising only 10 % of igneous rocks. Magnesium and calcium are very common in the earth's crust, with calcium the fifth-most-abundant element, and magnesium the eighth. Magnesium and calcium are found in many rocks and minerals: magnesium in carnallite, magnesite, dolomite and calcium in chalk, limestone, gypsum. Most strontium is found in the minerals celestite and strontianite. Barium is slightly less common, much of it in the mineral barite. Radium, being a decay product of uranium, is found in all uranium-bearing ores.




**FIREWORK**



Many alkaline and alkaline earth metals are used in creating colours, such as strontium and barium, are the colourful stars of a fireworks show. Combined with the element chlorine, barium sends up a green spark, calcium gives orange and lithium gives medium red. Strontium carbonate gives a bright red colour Nitrates of sodium gives orange, potassium and rubidium gives violet colour and caesium gives indigo colour. The burning "excites" the electrons, pushing them into higher than normal energy level; they release their extra energy as a colourful burst of light

The blue fireworks are the hardest to make, since the compound copper chloride breaks down in a hot flame. In recent years, fireworks experts have used magnalium- a mixture of the alkaline earth metal magnesium and aluminium - to boost all firework colours. Magnalium has made the blues brighter, but pyro technicians are still searching for a blue as brilliant as the red, green and yellow colours.

#### Electronic configuration

These elements have two electrons in the valence shell of their atoms, preceded by the noble gas configuration. Their general electronic configuration is written